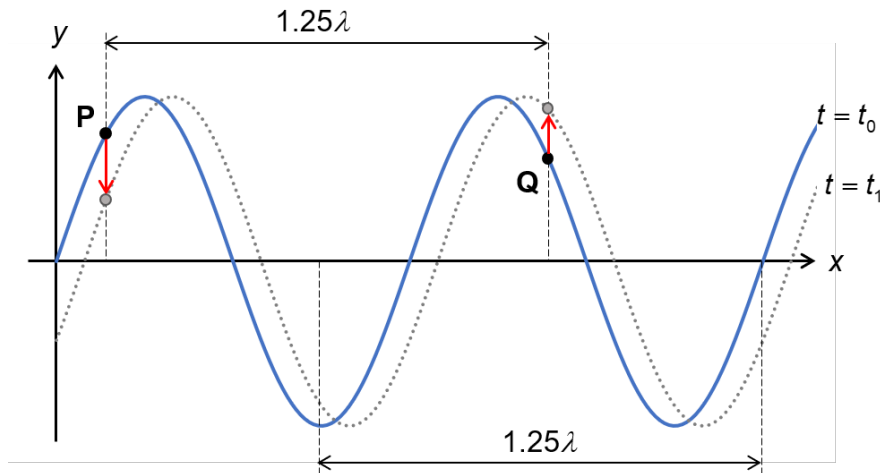


Tutorial 10 Wave Motion Suggested Solutions

D1 B



- D2 (a)** A transverse wave is one in which its particles oscillate in a direction perpendicular B1
to the direction of energy transfer,
while a longitudinal wave is one in which its particles oscillate in a direction parallel B1
to the direction of energy transfer.

(b) (i) $A = 0.50 \text{ Pa}$ A1

(ii) From graph,
 $T = 1190 - 350 = 840 \text{ } \mu\text{s}$ M1
 $f = \frac{1}{T} = \frac{1}{840 \times 10^{-6}} = 1190.48 = 1190 \text{ Hz}$ A1

(iii) $v = f\lambda$
 $340 = (1190)\lambda$ M1
 $\lambda = 0.28571 = 0.286 \text{ m}$ A1

- D3 (a)** From the graph, wavelength $\lambda = 0.80 \text{ m}$.
Since $v = f\lambda$,

$$f = \frac{v}{\lambda} = \frac{340}{0.80} = 425 \text{ Hz}$$

M1
A1

(b) Maximum speed of particle A $v_0 = \omega a = 2\pi f a$ M1

Total energy of the oscillation of particle A

$$E_T = \text{K.E.}_{\text{max}} = \frac{1}{2} m v_0^2 = 2m\pi^2 f^2 a^2$$

M1

Note: When K.E. is maximum, P.E. is zero.

(c) $E_T = 2m\pi^2 f^2 a^2$

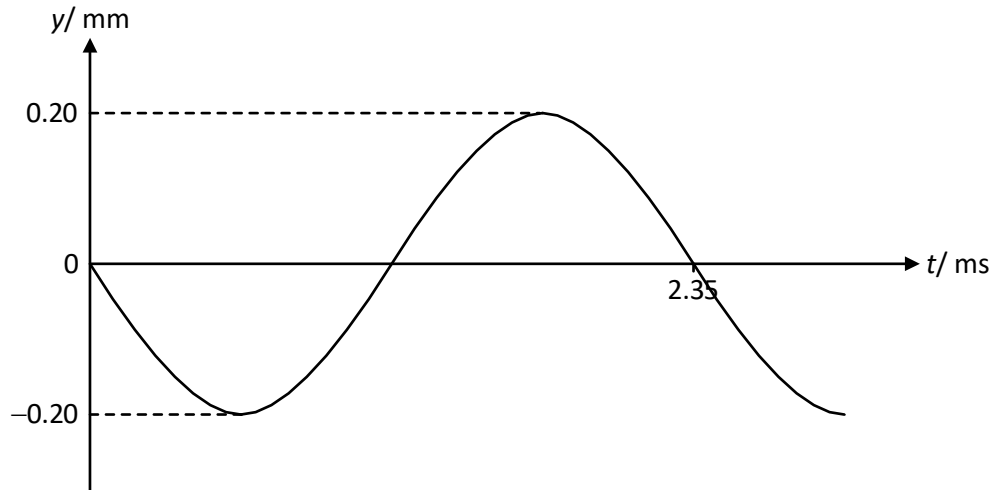
$$= 2\pi^2 \times 4.7 \times 10^{-26} \times 425^2 \times (0.20 \times 10^{-3})^2$$

M1

$$= 6.7029 \times 10^{-27} = 6.70 \times 10^{-27} \text{ J}$$

A1

(d)



Sinusoidal wave with axes labelled (amplitude and period); negative sine wave.
Note: The wave is travelling to the left.

B1

B1

(e) Let ϕ be the phase angle between A and B.

Motion of A is given by $y = -y_0 \sin \omega t$ while that of B is $y = -y_0 \sin(\omega t - \phi)$ where ϕ is positive.

At $t = 0$, for B

$$\sin(-\phi) = \frac{y}{-y_0} = \frac{0.10}{-0.20} = -0.50 \Rightarrow \phi = 30^\circ$$

M1

A1

Note: Particle B is lagging behind particle A by a phase of 30° .

D4 (a) $\omega = 200\pi = 2\pi f$

$$f = 100 \text{ Hz}$$

A1

(b) $v = f\lambda$

$$335 = 100\lambda \Rightarrow \lambda = 3.35 \text{ m}$$

$$\therefore \text{number of wavelengths} = \frac{500}{3.35} = 149.25$$

M1

A1

(c) $\frac{\Delta\phi}{2\pi} = \frac{x}{\lambda} = \frac{0.25\lambda}{\lambda}$

M1

$$\Delta\phi = \frac{\pi}{2} = 1.5708 = 1.57 \text{ rad}$$

A1

- D5** When the amplitude A of the wave increases, the intensity I at surface increases because $I \propto A^2$.

$$\frac{I_2}{I_1} = \left(\frac{A_2}{A_1} \right)^2 = \left(\frac{2A_1}{A_1} \right)^2 = 4$$

$$I_2 = 4I_1$$

M1

Since I is the rate of transfer of energy (P) per unit area normal to the direction of energy transfer of the wave, P will also increase.

$$I = \frac{P}{S} \Rightarrow P_1 = I_1 S_1 \quad \& \quad P_2 = I_2 S_2$$

M1

$$P_2 = (4I_1) \left(\frac{S_1}{2} \right)$$

$$= 2I_1 S_1$$

$$= 2P_1 = 2P$$

A1

- D6** Intensity is proportional to A^2 and $\frac{1}{r^2}$ (inverse square law for a point source).

$$\therefore A^2 \propto \frac{1}{r^2} \quad \text{OR} \quad A \propto \frac{1}{r} \Rightarrow \frac{A_2}{A_1} = \frac{r_1}{r_2}$$

M1

$$A_2 = \frac{r_1}{r_2} \times A_1$$

$$= \frac{3.0}{5.0} (1.0 \times 10^{-7})$$

$$= 6.00 \times 10^{-8} \text{ m}$$

M1

A1

- D7 (a)** Progressive

B1

- (b)** Transverse

B1

(c) time = $\frac{\text{distance}}{\text{speed}}$

$$= \frac{4.35 \times 10^{12}}{3.0 \times 10^8}$$

$$= 14500 \text{ s}$$

M1

A1

- (d) (i) Intensity of the signal 4.35×10^{12} m from the satellite (i.e. on Earth).

$$I_{\text{Earth}} = \frac{P_{\text{satellite}}}{4\pi r^2} = \frac{22.0}{4\pi (4.35 \times 10^{12})^2}$$

M1

Power received by the dish on Earth

$$= I_{\text{Earth}} S_{\text{dish}}$$

$$= \frac{22.0}{4\pi (4.35 \times 10^{12})^2} \times 260$$

M1

$$= 2.4055 \times 10^{-23} = 2.41 \times 10^{-23} \text{ W}$$

A1

- (ii) The power from the satellite is not radiated uniformly in all directions. It is directed towards the earth. B1

- D8 D** Energy is spread over the circumference of a circle ($2\pi r$) instead of the surface area of a sphere ($4\pi r^2$).

$$I \propto A^2 \text{ \& } I \propto \frac{1}{r} \Rightarrow \left(\frac{A_2}{A_1} \right)^2 = \frac{r_1}{r_2}$$

$$A_2 = \sqrt{\frac{r_1}{r_2}} \times A_1$$

$$= \sqrt{\frac{150}{1200}} (2.0)$$

$$= 0.70712 = 0.707 \text{ mm}$$

- D9 (a)** (i) A
(ii) I

- (iii) $I \propto A^2$ OR $I = kA^2$ where k is a constant

(b)

angle θ	amplitude	intensity
180°	A	I
90°	0	0
60°	$\frac{1}{2}A$	$\frac{1}{4}I$

D10 (a) Time taken for 5 complete waves
= 8 squares = $8 \times (1 \times 10^{-3}) = 8 \times 10^{-3}$ s M1

$$T = \frac{8 \times 10^{-3}}{5}$$

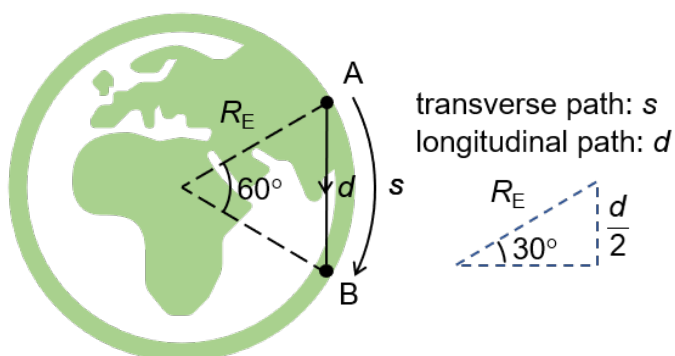
$$f = \frac{1}{T} = \frac{5}{8 \times 10^{-3}} = 625 \text{ Hz} \quad \text{A1}$$

(b) $v = f\lambda$
 $330 = 625\lambda$ M1

$$\lambda = 0.528 \text{ m}$$

$$0.5\lambda = 0.264 \text{ m} \quad \text{A1}$$

C1



For longitudinal waves,

$$\frac{d}{2} = R_E \sin 30^\circ$$

$$t_L = \frac{d}{v_L} = \frac{2R_E \sin 30^\circ}{v_L} = \frac{2(6.37 \times 10^6) \sin 30^\circ}{7800} = 816.6 \text{ s}$$

For transverse waves,

$$s = R_E \theta \text{ (in radians)}$$

$$t_T = \frac{s}{v_T} = \frac{(6.37 \times 10^6) \frac{\pi}{3}}{4500} = 1482.4 \text{ s}$$

Time difference between the arrivals of the waves at B = $t_T - t_L = 1482.4 - 816.6 = 666 \text{ s}$